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Q1. A Karachi-based startup is developing a cloud-based Hospital Management System (HMS). The system will: Support patient registration, billing, lab reports, and pharmacy integration. Integrate with third-party insurance APIs. Be deployed within 6 months. Initially, 3 hospitals but expected to scale nationwide. Requirements are partially known; doctors frequently change reporting formats. The client wants early usable modules.

The development team consists of 10 developers and 2 UI designers. The hospital management insists on seeing working modules every month. Now do the following:

- Select the most appropriate **software process model**. Justify your answer analytically.
- Explain why at least two other models would be less suitable.

Q2. A medium-sized software house in Lahore Has defined project management policies, maintains documentation standards, Uses defined coding standards, Has team hierarchy and formal reviews, Tracks project timelines but does not use quantitative metrics, Risk analysis is informal, No continuous process optimization activities. Identify the most appropriate CMM level. Justify using characteristics of that level. Explain why it does not qualify for the next higher level.

Q3. Give brief answers to following questions:

- In XP, a spike is created to explore technical uncertainty. Why should spike code not be used directly in production systems?
- A client requests major feature changes during an ongoing sprint. Should the Scrum team accept the changes immediately? Justify your answer using Scrum principles and sprint structure